

DWM

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4) Compare Data warehouse and Data mining.

Ans.	Data warehouse	Data Mining
	1) A Data warehouse is a central storage where data from different sources is collected and stored.	1) A Data mining is the process of analyzing large datasets to discover patterns and useful information.
	2) D.W is used for storing past (historical) data.	2) D.M. is used for predicting future trends.
	3) D.W focuses on data storage and management.	3) Data mining focuses on analysis and discovery.
	4) D.W works with clean and structured data.	4) D.M uses algorithms for prediction and classification. works with any type of data.
	5) D.W uses OLAP operations for reporting.	5) D.M uses algorithms for prediction and classification.
	6) D.W shows what has happened.	6) D.M explains why and what will happen.
	Example:- stores last 5 years of sales.	Example:- predict next year's sales.

6) Different between OLTP and OLAP operations.

Ans:-

OLTP

OLAP

1) OLTP stands for Online Transaction processing.

1) OLAP stands for online Analytical processing.

2) OLTP is used to run day-to-day business transactions.

2) OLAP is used to analyze data for decision-making.

3) OLTP stores current and detailed data.

3) OLAP stores historical and summarized data.

4) OLTP performs insert, update and delete operations.

4) OLAP performs roll-up, drill-down, slice, dice and pivot operations.

5) OLTP handles many small transactions at a time.

5) OLAP handles fewer but complex queries.

6) OLTP provides fast query processing for machine tasks.

6) OLAP provides slow but deep analysis of data.

7) Example:-

ATM withdrawal, railway booking.

7) Example:-

sales trend analysis, profit analysis.

10) compare Stars Schema and Snowflake Schema.

Ans:-

Stars Schema

Snowflake Schema

1) Stars Schema has a central fact table connected directly to dimension tables.

1) Snowflake Schema has a central fact table connected to normalized dimension tables.

2) Stars Schema dimension tables are denormalized.

2) Snowflake Schema dimension tables are normalized.

3) Stars Schema is easy to design and understand.

3) Snowflake Schema is more complex to design and understand.

4) Stars Schema queries run faster with fewer joins.

4) Snowflake Schema queries are slower due to many joins.

5) Stars Schema uses more storage space.

5) Snowflake Schema takes less storage space.

6) Stars Schema is less flexible for changes.

6) Snowflake Schema is more flexible for changes.

7) Stars Schema is best for small or medium data warehouse.

7) Snowflake Schema is better for large data warehouses.

8) Stars Schema is easy to understand for end-users.

8) Snowflake Schema is harder to understand for end-users.

9) Stars Schema supports quick reporting.

9) Snowflake Schema supports detailed analysis.

Q) Describe in details Data warehouse and explain its features, needs, application, benefits and approaches.

Ans:- A Data warehouse (DW) is a large, centralized repository that stores integrated data from multiple sources. It is designed for querying, analysis and decision-making, rather than daily transaction processing.

Features of D.W

- 1) Subject-oriented (e sales, customers, etc.)
- 2) Integrated (data from many sources)
- 3) Time-Variant (stores past data)
- 4) Non-Volatile (stable, not frequently changed).

Needs of D.W

- 1) To combine data from different department into a single view.
- 2) To improve data consistency and quality.
- 3) To support decision-making.
- 4) To analyze past trends.

Application of D.W

- 1) Retail :- Analyzing customer buying patterns.
- 2) Government :- Policy analysis, population statistics.
- 3) Banking
- 4) Telecom
- 5) Healthcare.

Benefits of D.W

- 1) Provide better decision-making.
- 2) Improve data quality and consistency.
- 3) Reduce time for data analysis.
- 4) Faster query results.

Approaches:-

- 1) Top-down
- 2) Bottom-up
- 3) Hybrid
- 4) Federated.

Q) Explain in details data warehouse Architecture.

Ans:- A Data warehouse Architecture defines how data is collected, stored, managed and made available for analysis. It includes various layers and components that work together to deliver clean and integrated data for decision-making.

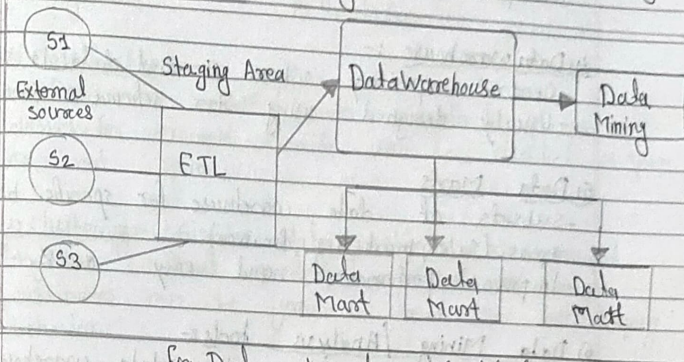


Fig-Data Warehouse Architecture.

Component of Data warehouse Architecture:-

1) Data Sources:-

- operational database, flat files, ERP, CRM, web data, etc.
- provide raw data to the warehouse.

2) ETL (Extract, Transform, Load) Process:-

- Extract data from multiple sources.
- Transform data by cleaning, filtering and converting into proper format.
- Load data into warehouse.

3) Staging Area:-

- Temporary storage used during ETL before loading into warehouse.
- Helps in data cleaning and transformation.

4) Data warehouse :-

- Central repository where processed data is stored.
- Usually designed using Stars schema or snowflake schema.

5) Data Marts

- Subsets of data warehouse for specific business areas (Sales, marketing, finance).
- Improve performance and easy access.

6) Data Mining / Analysis tools:-

- Tools that analyze the data warehouse to find patterns, trends and useful insights.
- supports OLAP operations (Roll-up, Drill-down, Slice, Dice, Pivot) and predictive models.

Types of Architectures:

- 1) Single-tier :- Rare, removes redundancy.
- 2) Two-tier :- Direct. Client-server connection.
- 3) Three-tier :- Most common (source → warehouse → analysis tools).

3) Describe in details Data Mining Techniques.

Ans:- Data mining is the process of discovering patterns, trends and useful information from large datasets. Different techniques are used depending on the type of data and the goal.

Data mining techniques are:-

- 1) Classification
- 2) Clustering
- 3) Regression
- 4) Association Rule Mining
- 5) Anomaly/Outliers Detection
- 6) Prediction

1) Classification:-

Data is divided into predefined groups or classes. For example, emails can be classified as spam or not spam.

2) Clustering:- Similar data objects are grouped together without predefined labels. For example, customers can be grouped based on their buying behaviors.

3) Regression:- This technique predicts continuous values. For example, it can be used to predict house price based on size and location.

4) Association Rule Mining:- It defines and finds relationships between items in a dataset. For example, if a customer buys bread, they are also likely to buy butter.

5) Anomaly Detection:- This is used to find unusual data that does not follow the normal pattern. For example, detecting fraud in credit card transactions.

6) Prediction:- Past data is used to forecast future outcomes. For example, predicting next month's sales.

5) Why the entity-relationship modeling techniques not suitable for the Data Warehouse? How is dimensional modeling different?

Ans. 1) Entity-Relationship (ER) modeling is designed for transactional systems (OLTP).

2) It focuses on normalization to reduce redundancy and ensures data consistency.

3) In a Data Warehouse (OLAP), the goal is fast querying and analysis, not frequent updates.

4) ER models are complex, involves many joins, and make queries slow.

5) Hence, ER modeling is not suitable for data warehouses where speed, simplicity and analysis are required.

How is Dimensional Modeling Different?

1) Dimensional modeling is designed for data warehouses.

2) It uses fact tables and dimension tables.

3) It focuses on denormalized denormalization for fast queries.

4) Schemas like star schema and snowflake schema are used.

5) Dimensional modeling is user-friendly, simple and support OLAP operations like Roll-up, Drill-down, slice and Dice.

ER Modeling

1) ER modeling design for OLTP systems (transactional).

2) It has normalized tables.

3) It is complex for users.

4) It is suitable for operations.

Dimensional Modeling

1) Dimensional modeling is designed for OLAP systems (analytical).

2) It has denormalized tables.

3) It is simple and user-friendly.

4) It is suitable for reporting.

Q. Explain different steps involved in Data Preprocessing.

Ans. Data preprocessing is the process of preparing raw data before applying data mining techniques. It improves the quality of data and ensures better results.

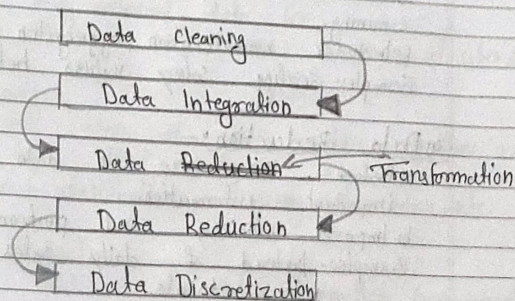


Fig:- Steps in Data Preprocessing.

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2) Data Cleaning:-

- a) Removes noise, errors and correct inconsistencies.
- b) Handles missing values, duplicate records and outlier outliers.

Example:- correcting wrong spelling in customer names.

3) Data Integration:-

- a) Combines data from multiple sources into one unified dataset.

- b) Resolves conflicts in naming, formats and units.

Example:- Merging sales data from two different branches.

4) Data Transformation:-

- a) Converts data into a suitable format for mining.

- b) Includes normalization, aggregation and generalization.

Example:- scaling salary values between 0 and 1.

5) Data Reduction:-

- a) Reduces the size of data while keeping important information.

- b) Improves efficiency and performance of data mining.

Example:- Instead of daily sales, storing monthly sales totals.

6) Data Discretization:-

converts continuous data into discrete categories.

Example:- Age values $\rightarrow$ "Young", "Middle-aged", "old".

Q.9 skip

Describe the steps involved in Data warehouse mining when viewed as a process of Knowledge Discovery.

Ans:- Knowledge Discovery in Databases (KDD) is the overall process of finding useful Knowledge from data. Data Mining is one important step inside KDD.

Steps in KDD:-

1) Data cleaning:- (write ans of preprocessing step).
same as same as previous question.

2) Data Integration:-

3) Data Transformation:- } also for them write same to same of (Q.7).

4) Data Selection:-

chooses only relevant data for analysis.

Example:- selecting only last 3 years of sales data.

5) Data Mining:-

- The core step of KDD where algorithms are applied to extract patterns.

- Techniques:- classification, clustering, Association, etc.

6) Pattern Evaluation:-

- Identifies valid, useful and interesting pattern from mined result.

- Removes irrelevant or redundant findings.

Example:- Ignore trivial rules, keep meaningful ones.

7) Knowledge Presentation:-

- Final step where results are shown using visual visualization or reports.

Example:- Graphs, charts, dashboards.

11) Explain different Data Visualization Techniques.

Ans:- Data visualization is the process of representing data in a graphical or visual format so that patterns, trends and insights can be easily understood. It helps decision-makers quickly analyze data.

1) Bar chart:-

- a) Represents data using rectangular bars.
- b) Useful for comparing quantities across categories.
- c) Bars can be vertical or horizontal.

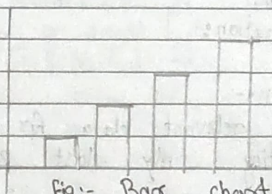


Fig:- Bar chart

2) Line chart:-

- a) Display data points connected by lines.
- b) Used for showing trends over time.

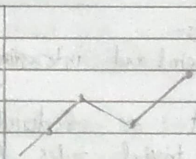


Fig Line chart

Q-12 skip

3) Histogram:-

- a) Similar to a bar chart but represent frequency distribution of continuous data.
- b) Useful for identifying patterns or distributions.

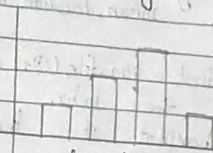


Fig:- Histogram

4) Scatter Plot:-

- a) Shows data points plotted on X and Y axes.
- b) Useful for finding relationships or correlations.

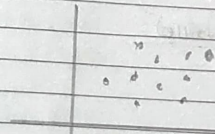


Fig:- Scatter plot

5) Bubble chart:-

- a) Similar to scatter plot but uses bubble size to show an additional variable.
- b) Useful for multivariable comparisons.

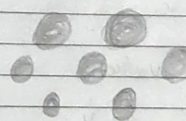


Fig:- Bubble chart.

- 14) Suppose that the data for analysis include the attribute salary. We have the following values for salary (in thousands of dollars), show in increasing order: 30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, 110.
- What are the mean, median, mode and midrange of the data.
 - Find the first quartile (Q_1) and the third quartile (Q_3) of the data.
 - Show the boxplot of the data.

Ans. We have salary data (in thousands):-
30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, 110

i) Mean, Median, Mode, Midrange

$$a) \text{Mean} = \frac{\text{Sum of values}}{\text{(count)}}$$

$$= \frac{696}{12}$$

$$\text{Mean} = 58$$

b) Median (middle value)

$$\text{Median} = \text{average of 6th and 7th values}$$

$$= \frac{(52+56)}{2}$$

$$\text{Median} = 54$$

c) Mode (most frequent value)

$$\text{Mode} = 52 \text{ and } 70 \text{ (both appear twice)}$$

d) Midrange

$$\text{Midrange} = \frac{(\text{min} + \text{max})}{2}$$

$$= \frac{(30 + 110)}{2}$$

$$\text{Midrange} = 70$$

ii) Quartiles:-

$$Q_1 = \text{median of lower half } \{30, 36, 47, 50, 52, 52\}$$

$$= \frac{(47 + 50)}{2}$$

$$Q_1 = 48.5$$

$$Q_3 = \text{median of upper half } \{56, 60, 63, 70, 70, 110\}$$

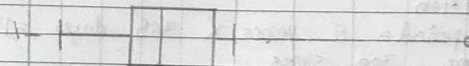
$$= \frac{(63 + 70)}{2}$$

$$Q_3 = 66.5$$

iii) Box plot

$$\text{Min} = 30, Q_1 = 48.5, \text{median} = 54, Q_3 = 66.5, \text{Max} = 110$$

salary Boxplot



30 40 50 60 70 80 90 100 110

salary (in thousands)

13) Ans:- 1) Store schema.

Product		STORE
Product-key		Store-key
Product-name		Store-name
SKU Number		Store-ID
Branch name		Address
Product category		city
weight		state
Package type		zip
shelf level		District
shelf width		Manager
shelf depth		Service Type

TIME		PROMOTION
Time-key		Promotion-key
Date		Promotion-name
Day of week		Promotion-Type
month		Promotion cost
years		Start Date
week number		END Date
month number		Media type

ii) calculation

Time period = 5 years $\times$ 365 days = 1825

There are 300 stores

Each store daily sale = 4000

Promotion = 1

Maximum number of fact table records = $1825 \times 300 \times 4000 \times 1$

= 219,600,000
= 2 billion